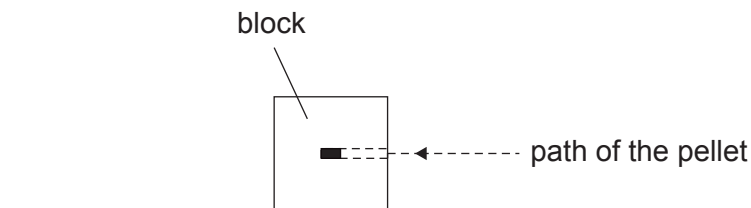


PAPER 1 / QUESTION 77

4 November 2021 / TZ0 / HL Paper 1 / original question 6 / 8821-6501

77. A block rests on a frictionless horizontal surface. An air rifle pellet is fired horizontally into the block and remains embedded in the block.



What happens to the total kinetic energy and to the total momentum of the block and pellet system as a result of the collision?

	Total kinetic energy	Total momentum
A.	no change	no change
B.	no change	decreases
C.	decreases	no change
D.	decreases	decreases